

(Time: 2 hours)

[Total Marks: 60]

- N. B.: (1) All questions are compulsory.
(2) Make suitable assumptions wherever necessary and state the assumptions made.
(3) Answers to the same question must be written together.
(4) Numbers to the right indicate marks.
(5) Draw neat labeled diagrams wherever necessary.
(6) Use of Non-programmable calculator is allowed.

1. Attempt any two of the following: 12

- a. What is frameworks and Explain different types of framework?
- b. How can you create new ASP.NET Core projects using development tools?
- c. Why is unit testing important in the development of ASP.NET Core applications?
- d. What is .net SDK? Explain different dotnet commands.

2. Attempt any two of the following: 12

- a. How do you display a list of products in a Blazor application? What data source might you use for the same?
- b. Discuss the importance of styling in a web development and how can we style a content in a Blazor application.
- c. Explain the purpose of shopping cart in an e-commerce application.
- d. Describe the process of adding catalogue management features to a Blazor application.

3. Attempt any two of the following: 12

- a. How can you apply basic Authorization policy to restrict access to a specific part of ASP.NET Core application?
- b. How can cookies and sessions be utilized for user state management in ASP.NET Core?
- c. Discuss the dependency injection features that can enhance routing functionality in ASP.NET Core.
- d. How can custom middleware be created and integrated into ASP.NET Core application?

4. Attempt any two of the following: 12

- a. Why is caching data important in a web application, and what are the advantages of doing so?
- b. How can you integrate a CSS framework into your project for styling purposes?
- c. What are the best practices for exploring and testing web services during development?
- d. What is the HTTP PATCH method? Why is it important for web service development?

5. Attempt any two of the following: 12

- a. Explain the process of modifying a Razor view in ASP.NET MVC.
- b. When and why would you use the ViewBag in a Razor view?
- c. Describe the different types of view component results.
- d. Describe the process of creating a custom tag helper.